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North America 2025

Getting Up To Date With Docsy: The Kubernetes Docs upgrade in progress

Divya Mohan, SUSE

Natali Vlatko, Cisco

Rey Lejano, Red Hat

Sayak Mukhopadhyay, Gemini Solutions

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Introduction



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Speakers



Divya Mohan, SUSE

- Kubernetes maintainer
- Kubernetes SIG Docs co-chair
- KubeCon India co-chair
- DRF Steering Committee



Natali Vlatko, Cisco

- Kubernetes maintainer
- Kubernetes SIG Docs co-chair
- TODO Group Steering Committee
- CNCF Maintainer Summit Program Co-chair



Rey Lejano, Red Hat

- Kubernetes maintainer
- Kubernetes SIG Docs co-chair
- Kubernetes SIG Security's Third-Party Audit subproject lead



Sayak Mukhopadhyay, Gemini Solutions

- Kubernetes SIG Docs approver
- Kubernetes SIG ContribEx contributor

Today's Roadmap



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Brief agenda

- The upgrade project
- Why we're upgrading
- How we're approaching it
- Roadmap & current status
- How you can get involved

Setting the Stage



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Why Docs Matter



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Essential for onboarding

Docs help new users and contributors get started quickly.

Supports rapid innovation

Clear docs enable collaboration and faster development in open source.

Empowers the community

Up-to-date docs let users deploy, manage, and troubleshoot Kubernetes confidently.

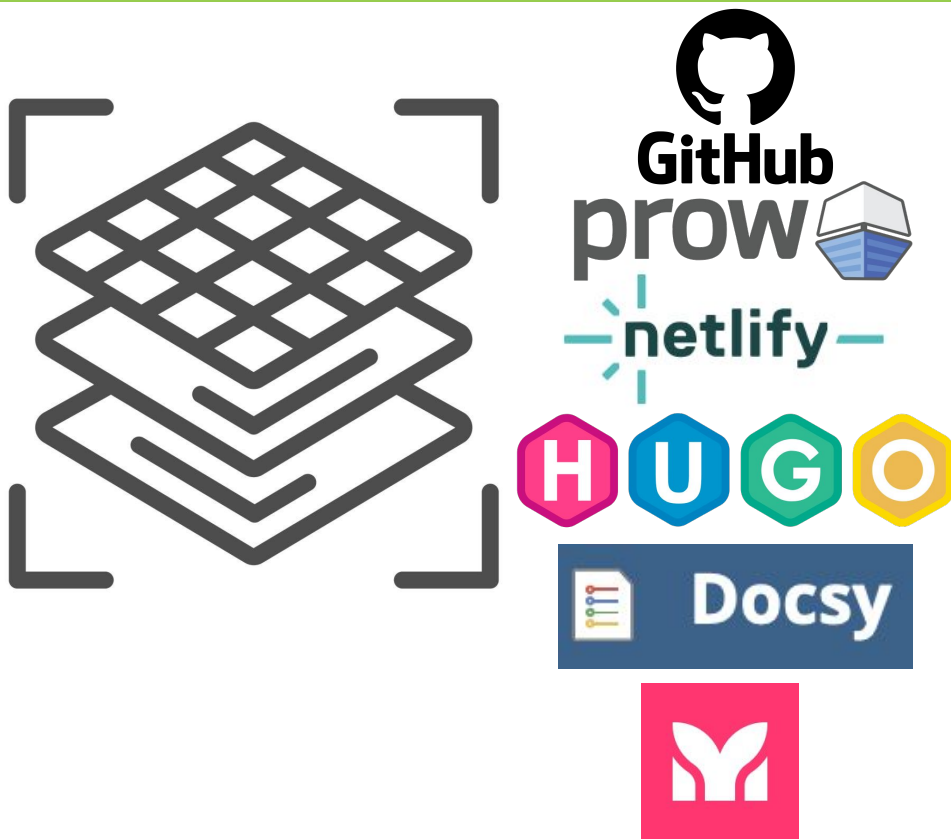
Reduces support burden

Good docs mean fewer support requests and more self-service.

Drives adoption

High-quality documentation is key to Kubernetes' global growth and success.

kubernetes.io Tech Stack



GitHub - source control

Prow - Kubernetes CI/CD system

Netlify - hosting platform, builds via GitHub

Hugo - static site generator and templating (extends Go templating)

Docsy - Hugo theme for technical documentation

Mermaid.js - JavaScript-based tool to dynamically create diagrams from markdown

Why Staying Up to Date Matters



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Security

Latest versions patch vulnerabilities and keep the site safe for users and contributors.

Compatibility

New Docsy releases are designed to work with recent Hugo versions; outdated setups can break site builds and features.

Modern Features

Upgrades unlock features like dark mode, improved navigation, and better accessibility.

Performance

Newer Hugo versions offer faster builds and better resource management.

Community Contributions

Up-to-date tools make it easier for contributors to onboard and help, reducing friction and errors.

Future-Proofing

Staying current ensures the docs site can adopt new improvements and avoid technical debt.

The Upgrade Initiative



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What's changing?



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Migrations

- Update to the latest Docsy
- Update to the latest compatible Hugo
- Make the setup as vanilla as possible

Goals

- Security
- Features
- Maintainability
- Future-proofing

The Backstory



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Early Years - Jekyll-Based (2014-2018)



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The original Kubernetes documentation site was built using Jekyll.

Jekyll enabled rapid initial development but became limiting as the project and documentation needs grew.

Migration to Hugo (Mid-2018)

In mid-2018, the site migrated to Hugo, a faster and more flexible static site generator.

Hugo improved build times, supported complex content structures, and made it easier to manage multiple documentation versions.

Adoption of Docsy (Mid-2020)

In mid-2020, the Docsy theme was adopted to modernize the site's look and feel.

Docsy brought better navigation, improved accessibility, and support for localization and versioning.

Theme Updates/Technical Debt (2020-2024)



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The Docsy theme was updated regularly until late 2024, reaching version 0.3.0.

After 0.3.0, updates slowed, and the site began to lag behind Hugo releases, leading to security and compatibility concerns.

Why Upgrade Now?



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The site needs to keep pace with Hugo and Docsy updates to patch security issues, unlock new features, and maintain a high-quality user experience.

Upgrading reduces technical debt and ensures the documentation remains accessible and easy to contribute to.

Our Approach



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Approach

Make small PRs

Update one layout or shortcode at a time

Clean things up while you are at it

Try to think one version ahead

Rebase to keep your working branch up-to-date

Process



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Audit current site and dependencies.

- Review the existing documentation site, its structure, and all dependencies.
- Identify customizations, outdated components, and areas that may need special attention.

Compare current and latest Docsy versions.

- Download both the current and target (latest) versions of the Docsy theme.
- Use diff tools to compare templates, layouts, and configuration files between versions.
- Note any breaking changes or deprecated features that could impact the site.

Incrementally update templates and layouts.

- Update templates and layouts one at a time, starting with the most critical or outdated.
- Make small, focused changes to minimize risk and simplify troubleshooting.
- Clean up code and remove deprecated elements as you go

Test changes and gather feedback.

- After each update, thoroughly test the site locally and in staging environments.
- Solicit feedback from contributors and users to catch issues early.
- Adjust and refine updates based on real-world usage and feedback.

Update Docsy dependency version.

- Update the Docsy dependency in the project configuration.
- Ensure the site builds and functions correctly with the new version

Where We Stand



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Current Status



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Progress So Far

- Completed initial audit of site and dependencies.
- Compared current and latest Docsy versions.
- Identified key areas needing updates.

Ongoing Work

- Incremental updates to templates and layouts are in progress.
- Testing and feedback collection underway.
- Small, focused PRs being submitted and reviewed.

The Road Ahead



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Roadmap



Join the Effort



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Join Us!

Review and test the
upgraded
documentation site.

Contribute to
template and layout
updates.

Submit feedback,
bug reports, or
suggestions.

Help review and
merge pull requests.

Join discussions in
SIG Docs meetings
and community
channels.

Every
contribution—big or
small—helps
improve Kubernetes
docs for everyone!

Connect with us

- [#sig-docs](#) in the [Kubernetes Slack](#)
- <https://bit.ly/k8s-sig-docs>
- SIG Docs meet bi-weekly on Tuesdays at 17:30 UTC
- SIG Docs APAC-timezone friendly meeting on the Wednesday after the fourth Tuesday of the month at 5:30 UTC
- New Contributor Meet and Greet on the first Tuesday of the month at 10:30 UTC
- Localization subproject meets on the first Monday of the month at 15:00 UTC
- Mailing list: <https://groups.google.com/g/kubernetes-sig-docs/>

Thank You



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